

Dissecting Dota

A statistical analysis of DotA 2

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1 Introduction

Defense of the Ancients (DotA), is a competitive multiplayer video game. Originally developed as a custom map for Warcraft III, it has evolved into a standalone title with a massive global player base. Two teams of five players each engage in strategic battles, navigating a dynamic map and competing against both minions and enemy players. The ultimate goal is to destroy the enemy team’s Ancient, a crucial structure within their base.

In DotA, each player chooses a hero, with a unique set of skills, contributing to the diverse team strategies. Players assume distinct roles within the team, namely "Carry", "Mid", "Offlaner", "Soft support" and "Hard support", commonly referred to as "Position 1" through "Position 5". The battle unfolds across three lanes - Top, Mid and Bottom - with each serving as a critical pathway for advancing towards enemy base. A crucial aspect of gameplay is accruing gold and experience points, achieved by defeating both minions and enemy heroes. This process of leveling up heroes and acquired items is central to gaining a tactical advantage. Additionally, players aim to destroy enemy towers, the defensive structures in each lane, to establish map control. Successfully destroying these towers not only weakens the enemy’s defenses but also positions the team more favorably to seize control of the game.

Our study aims to explore different factors that influence a team’s chances of winning a game. The initial focus is on the impact of resource distribution within a team on their likelihood of winning. This aspect of the investigation revolves around analyzing the networth ratio as a key parameter, with the outcome of each game (win or loss) serving as the response variable. We further explore the influence of other factors, such as the networth of individual players, kills secured by a player, and towers lost by the team on the outcome of a match.

2 Summary of the dataset

The dataset sources from Stratz[1] comprises of 8,250 matches. A random walk methodology with restrictions ensured one game per player, all within the same skill bracket. Post-collection, we preprocessed the data to remove matches with outliers, NaN values and one-sided games. The final dataset, reduced to 7500 matches, includes detailed metrics like player roles, hero choices, resource metrics at various intervals and match outcomes. Match outcome and hero type are categorical variables, while the remaining metrics are numerical values. For analysis, the matches were partitioned into two sets to analyze the winning and losing teams separately, ultimately focusing on 4,500 matches after considering hero restrictions.

The resource metrics in game are influenced by the duration of the game. Therefore, the analysis focused on the net worth of the five team positions at 10 and 20 minutes into the game. A crucial aspect was managing the extensive pool of 120 heroes. To ensure a practical and effective analysis, the study limited its focus to 40 heroes, selected based on their frequency of appearance in the dataset. This approach prioritized the most commonly chosen heroes, providing a balanced representation of popular gameplay strategies.

3 Identifying the key factors

To identify key factors affecting game outcomes, our approach involved a thorough analysis of the dataset. We utilized various visualization techniques and statistical metrics to understand the influence of different variables, such as player roles, hero choices, and resource metrics, on match results. (Appendix A for data visualizations)

We eliminated factors with high skewness, direct correlation or high multicollinearity to refine the model. With results from histograms, boxplots, and a correlation matrix, we selected the most relevant variables: net worths of the five team positions, towers lost, and match outcomes at 10 or 20 minutes. These variables are not only statistically significant but also have practical relevance in the game context.

Additionally, we conducted AIC tests with forward model selection and interaction tests to validate selected parameters. The AIC tests confirmed the significance of the selected variables, while the interaction tests revealed no significant interactions between the variables. The results of these tests are included in Appendix B and C respectively.

4 Methods

For our analysis, logistic regression was chosen as the statistical model due to its effectiveness in handling binary response variable - in this case, match outcomes categorized as wins (1) or losses (0). The model is fit using maximum likelihood estimation for 10 and 20 minute net worths separately. This model is ideal for exploring relationships between independent variables, like resource distribution, and a binary outcome.

There are two main questions that we ask in our analysis:

1. Is the distribution of in-game resources among players a determining factor in the outcome of a match?
2. Which factors are most crucial in determining a team's likelihood of securing a Win?

4.1 Model 1

To approach the first question, we use the following logistic regression model:

$$\text{logit}(p) = \beta_0 + \beta_1 \times \text{Ratio}_1 + \beta_2 \times \text{Ratio}_2 + \beta_3 \times \text{Ratio}_3 + \beta_4 \times \text{Ratio}_4$$

$$p = \frac{1}{1 + e^{-\text{logit}(p)}}$$

- where p is the probability of a team winning the match
- Ratio_1 through Ratio_4 are the ratios of net worths of the Position_1 through Position_4 respectively (Position_5 is excluded to maintain linear independence - as they sum to 1)
- β_0 through β_4 are the coefficients of the model

4.2 Model 2

To approach the second question, we used the following logistic regression model after considering several parameters and running diagnostic tests. (Appendix D)

$$\text{logit}(p) = \beta_0 + \beta_1 \times \text{NW}_1 + \beta_2 \times \text{NW}_2 + \beta_3 \times \text{NW}_3 + \beta_4 \times \text{NW}_4 + \beta_5 \times \text{NW}_5 + \beta_6 \times \text{Towers}$$

$$p = \frac{1}{1 + e^{-\text{logit}(p)}}$$

- where p is the probability of a team winning the match
- NW_1 through NW_5 are the absolute net worths of the five team positions
- Towers is the number of towers lost
- β_0 through β_6 are the coefficients of the model

4.3 Model selection

To find an appropriate model, we ran diagnostic tests on every possible parameter combination. Our first step was to eliminate models with residual deviations greater than 0.3 from the horizontal line. Following this residual-based filtering, we added an additional criterion by removing models with low Area Under the Curve (AUC) values.

Following that, we prioritized ten models based on their Akaike Information Criterion (AIC) scores. To reach a final decision, we manually examined these ten models, choosing the one that exhibited the most favorable residual vs. probability graph.

5 Results

5.1 Model 1

The analysis outcomes primarily stem from the likelihood test and Wald tests conducted on the model parameters. In the likelihood test, the reduced model consists of only the intercept as a parameter, serving as a baseline for comparison.

- **Null Hypothesis:** The networth ratios of the five team positions do not affect the outcome of the match.
 $H_0 : \beta_1 = \beta_2 = \dots = \beta_4 = 0$
- **Alternate Hypothesis:** The networth ratios of at least one position affects the outcome of the match.
 $H_1 : \beta_i \neq 0$ for at least one i

Likelihood test: Full model vs Reduced model	Result
Test statistic (chi-squared)	-1128
p-value	< 0.001

The p-value is less than 0.001, which shows that not all parameters are zero.

We performed a Wald test for each individual parameter to assess its significance. The findings suggest that the distribution of resources among different roles does not influence the probability of winning where the p-Values correspond to the result of the Wald test on the parameter.

Parameter	Value	p-Value
Position 1 Net-worth Ratio	-2.38	0.66
Position 2 Net-worth Ratio	-1.5	0.78
Position 3 Net-worth Ratio	-3.99	0.47
Position 4 Net-worth Ratio	-4.48	0.42

5.2 Model 2

Similar to model 1, the results of the analysis are the results of likelihood test and Wald tests.

- **Null Hypothesis:** The absolute networth of the five team positions and the number of towers lost do not affect the outcome of the match.
 $H_0 : \beta_1 = \beta_2 = \dots = \beta_6 = 0$
- **Alternate Hypothesis:** At least one of the parameters affects the outcome of the match.
 $H_1 : \beta_i \neq 0$ for at least one i

Likelihood test: Full model vs Reduced model	Result
Test statistic (chi-squared)	1782
p-value	< 0.001

Likelihood test indicates that not all parameters are zero.

Results of Wald test on each parameter:

Parameter	Value	p-Value
Position 1 Net-worth	3.45	< 0.001
Position 2 Net-worth	3.35	< 0.001
Position 3 Net-worth	3.16	< 0.001
Position 4 Net-worth	1.12	< 0.001
Position 5 Net-worth	0.99	< 0.001
Towers lost	-1.21	< 0.001

The result indicates that all parameters are significant.

Although the coefficients were different, there was no change in hypothesis test results for both models at 10 mins and 20 mins.

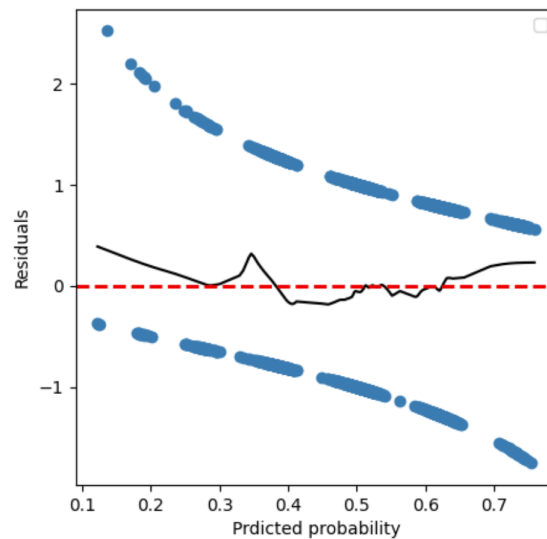
6 Model diagnostic

6.1 Verifying assumptions

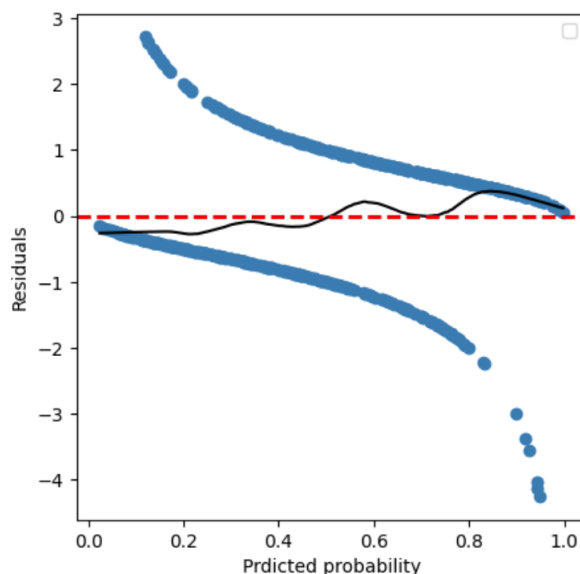
- Independence of the observations is maintained in the data collection.
- The VIF score of each parameter under consideration in both models is around 1, thus confirming no multicollinearity.

6.2 Residual Test

The pearson residual vs estimated probability shows a curve close to 0 for model 1. This shows that the model is not a good fit with the data. We attribute this to not including player heroes as parameters in the model.



Although the pearson residual vs estimated probability for model 2 shows a curve close to 0, the pattern in the graph shows that the model did not fit the data well. Similar to model 1, we attribute this to missing parameters in our model.



7 Discussion

Our analysis indicates that without considering the heroes chosen by players, the distribution of net worth doesn't seem to affect a team's chance of winning. However, individual net worth positively influences the probability of winning, while the number of towers lost by a team has a negative impact. These patterns hold true across all three game durations we examined.

A significant drawback of our study is not including hero choices as a factor in our models. Some heroes perform differently against specific opponents, and hero choices might interact with net worth, affecting our findings. This omission weakens the reliability of our results.

Both of our models display signs of overdispersion or noticeable patterns in residual plots. Exploring alternatives to logistic regression or including more parameters might help improve our results.

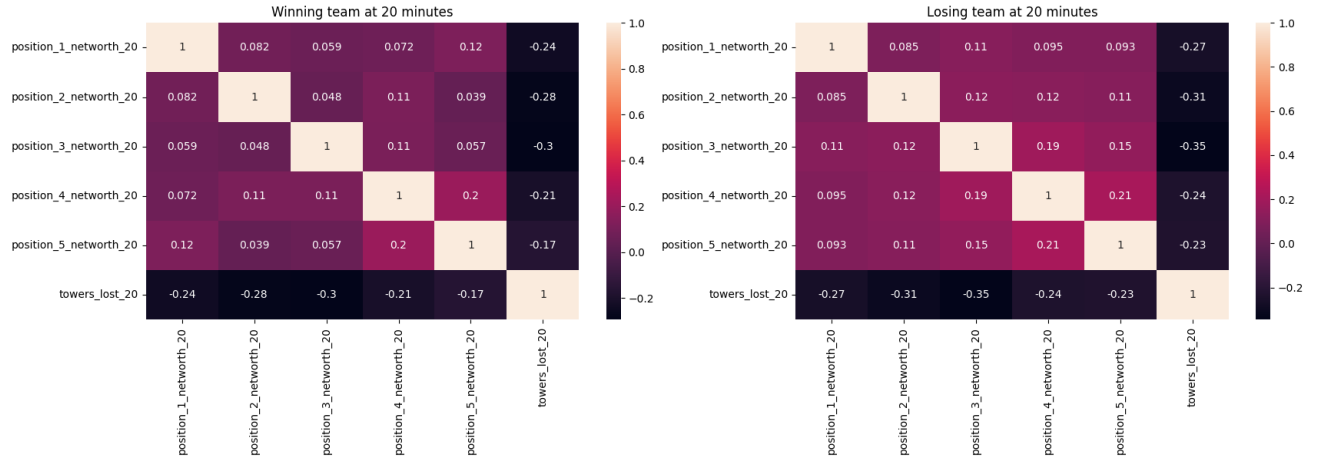
Additionally, we couldn't confirm that our models meet the assumption of linearity. This means our results could change if there are nonlinear factors affecting the model. It's important to recognize this limitation and consider potential adjustments in future analyses.

8 Contribution

- Sathkrith: Hypothesis Testing, Model selection, Model validation, Verification of assumptions, Report.
- Madhukara: Data collection, Data pre-processing, Exploratory data analysis, Model selection, Report.

9 Appendix

A Data visualization to identify key factors



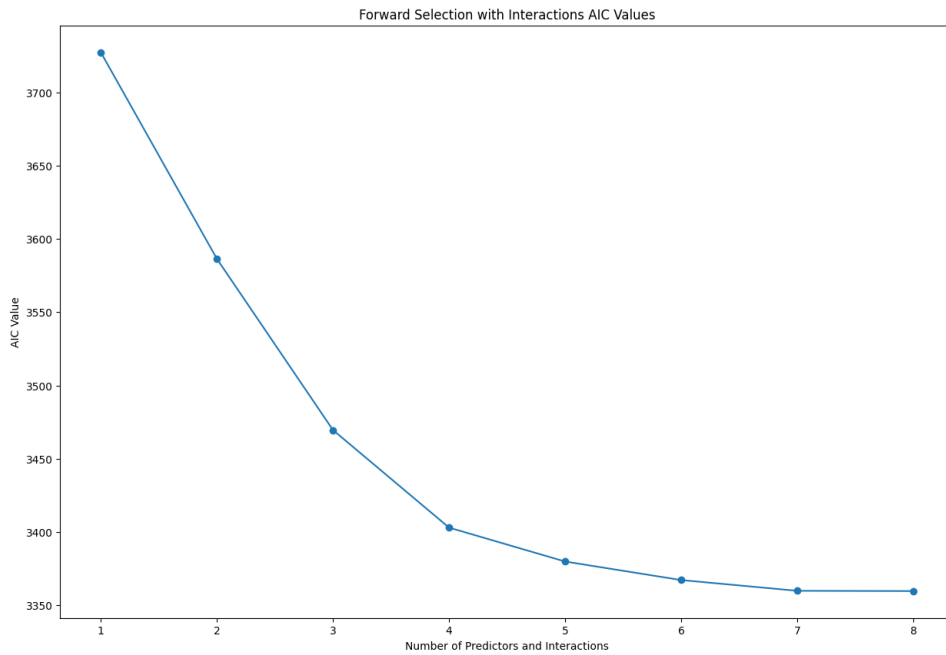
B AIC test with forward model selection

Akaike Information Criterion is a measure used for statistical model selection, including logistic regression. It balances the goodness of fit and model complexity, aiming to identify the model that best explains the data while penalizing for including additional parameters that may not be necessary.

It is calculated as follows:

$$AIC = -2 \times \text{loglikelihood} + 2 \times \text{number_of_parameters}$$

Lower AIC values indicate a better trade-off between fit and complexity.



C Interaction tests

Interaction variables

- Model with interaction = Pos1 + Pos2 + Pos3 + Pos4 + Pos5 + Towers lost + (Pos1 * Pos5) + (Pos3 * Pos4) + (Pos1 * Pos4)
- Conclusion: Interactions do not provide a better fit than the base model

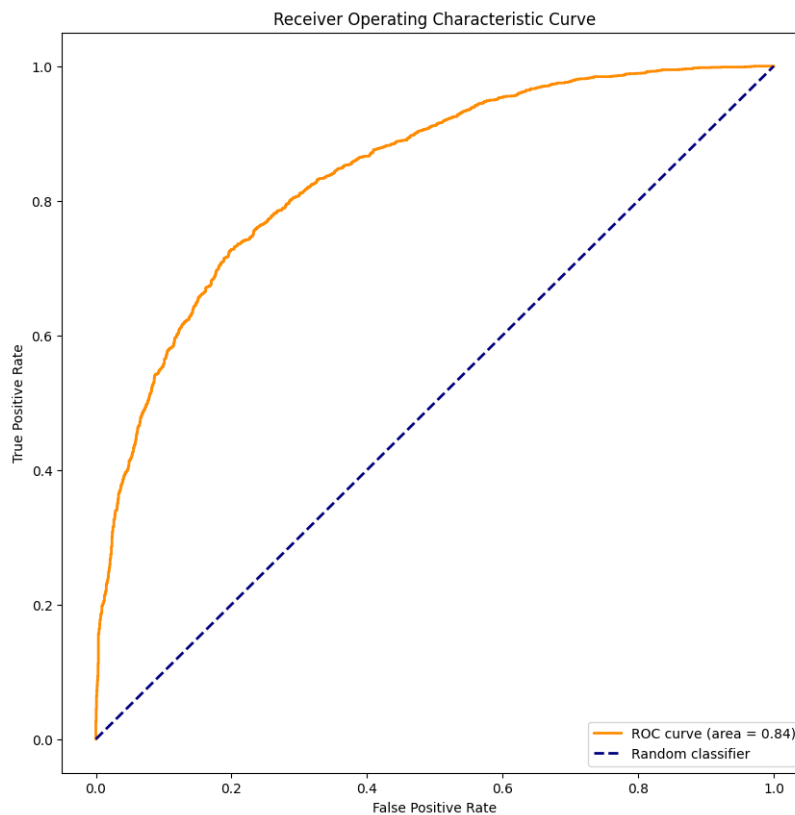
Interaction model	Position 1 network	Position 2 network	Position 3 network	Position 4 network	Position 5 network	Towers lost	Pos1xPos5	Pos3xPos5	Pos1xPos4
Coefficients	2931	3.35	~0	~0	8731	-1.21	-8728	12145	5800
P-value	0.852	<0.001	0.435	0.009	0.06	<0.001	0.06	0.435	0.71

D Diagnostic tests

ROC Curve

The Receiver Operating Characteristic (ROC) curve is a graphical representation used in binary classification to assess and visualize the performance of a predictive model. It illustrates the trade-off between the true positive rate (sensitivity) and the false positive rate (1-specificity) across different threshold values.

The Area Under the ROC Curve (AUC) is a scalar value that quantifies the overall performance of the model. A higher AUC indicates better discrimination between positive and negative instances.



E Hypothesis testing for Logistic regression

We use likelihood ratio test to compare the full and reduced model to determine if all parameters are zero.

- **Null Hypothesis:** All parameters are zero (not significant)
 $H_0 : \beta_1 = \beta_2 = \dots = \beta_n = 0$
- **Alternate Hypothesis:** Not all parameters are zero (some are significant)
 $H_1 : \beta_i \neq 0$ for at least one i

The actual test statistic for the likelihood ratio test, denoted by G^2 , is:

$$G^2 = -2 \log \frac{L(R)}{L(F)} = -2[\log_e L(R) - \log_e L(F)]$$

If

$$G^2 \leq \chi^2(1 - \alpha; df_r - df_f), \text{ conclude } H_0$$

$$G^2 > \chi^2(1 - \alpha; df_r - df_f), \text{ conclude } H_a$$

To test individual parameters, we use the Wald test with Bonferroni correction.

$$H_0 : \beta_1 = 0$$

$$H_a : \beta_1 \neq 0$$

Then test statistic z^* is given by:

$$z^* = \frac{\beta_k}{s(\beta_k)}$$

$$|z^*| \leq z(1 - \frac{\alpha}{2g}), \text{ conclude } H_0$$

$$|z^*| > z(1 - \frac{\alpha}{2g}), \text{ conclude } H_a$$

References

- [1] Stratz - dota 2 stats. <https://stratz.com>. Accessed: 2023-12.